

Hackathon Day 2

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AI4Imaging Summer School

7 - 11 September 2026



MEDICAL UNIVERSITY
OF VIENNA

CIR

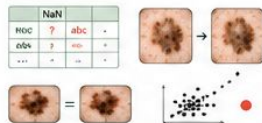
Hackathon Day 1
AI4Imaging Summer School

1 Data Curation

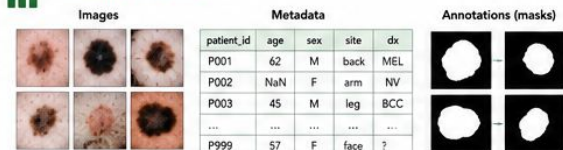


Simulated real clinical dataset (subset of ISIC)

- Messy data on purpose: missing values, duplicates, inconsistent types, outliers, ...



Visualize the data



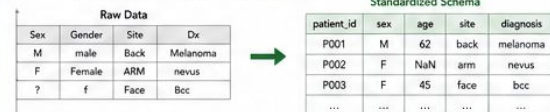
Explore & assess quality

- Get an idea of how the data looks, how organized it is, and how good the quality is.
- What have you noticed about the images?
- What issues are you expecting to arise?
- What do you consider to be basic requirements each image + metadata pair?



Data Transformation & Standardization

- Reshape, rename, or map values to match a common schema or controlled vocabulary



Define your own splits: Train / Val / Test

- Be careful with what unit you use to do the split. Would you divide images, lesions or people?
- What issues do you see with each of these?

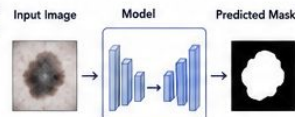


2 Segmentation



Use a segmentation model

- Goal: predict lesion mask from dermoscopic image



Train the model

- Train the model on the dataset
- Data augmentation & preprocessing (color jitter, flipping, resize, etc.)



Performance metrics

- Provide relevant metrics for validation and test set

Metric	Val Set	Test Set
Dice Score	0.86	0.84
IoU (Jaccard)	0.76	0.74
Precision	0.87	0.85
Recall	0.87	0.84
Hausdorff (mm)	7.2	7.8
...	...	...



Target segmentation Dice Score: above 80%

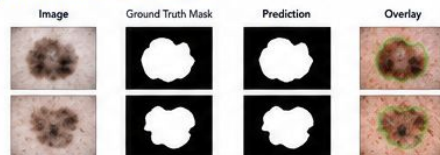


Suggested architectures

- Examples:



Qualitative results (examples)

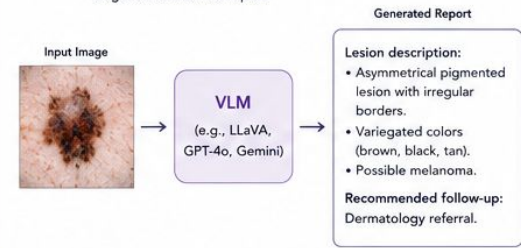


3 Generating lesion reports with VLM (optional)



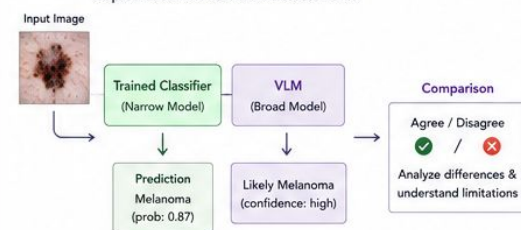
Use a general purpose VLM

- Provide image(s) and prompt to a Vision Language Model to generate a clinical report



Compare with trained classifier (optional)

- Train a classifier on the curated data and compare its predictions with the VLM's assessments



Report evaluation (optional)

- If ground-truth reports / labels exist, evaluate the quality

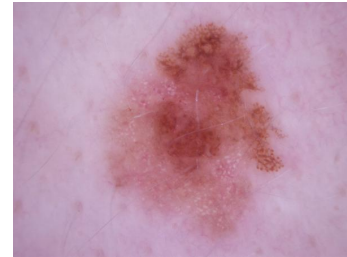
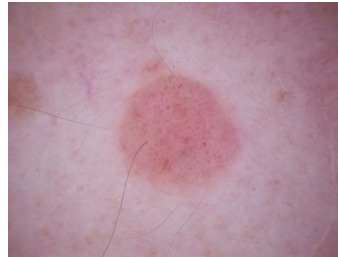
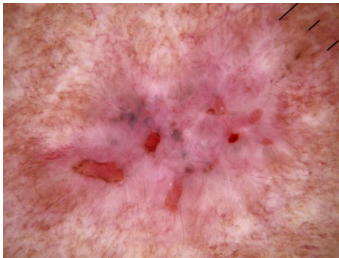


Data Curation

Task 1

ISIC dataset

Demoscopic and clinical skin lesion images collected for dermatology and skin cancer research.



ISIC dataset

> 1.000.000 images

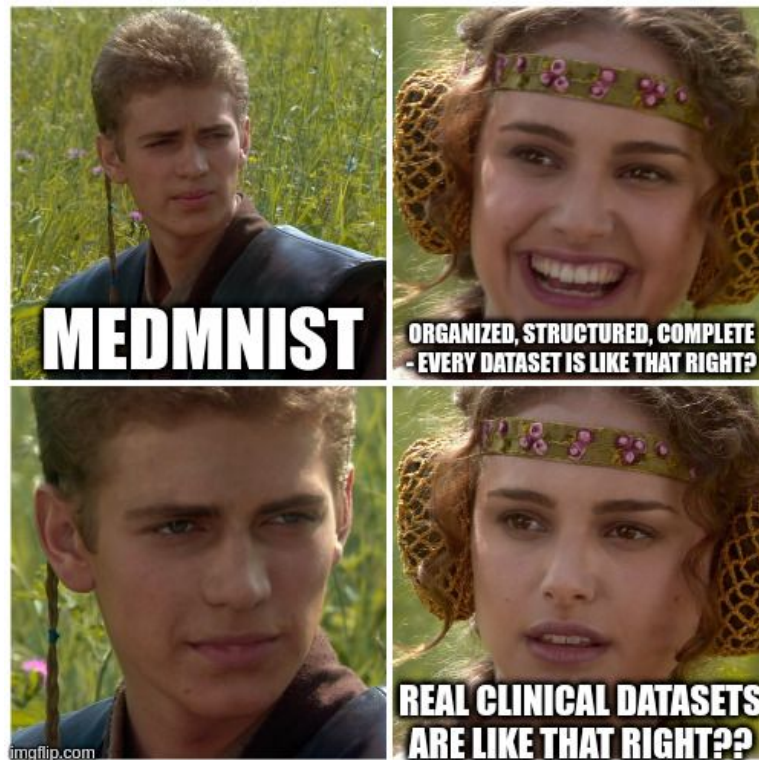
53 datasets

Subset

17k images

Your tasks

- Clean up the dataset
 - set it up for AI training/testing
- Build a lesion segmentation
- Generate a report using VLMs



Setup

Open the terminal in your SageMaker account

type

```
aws s3 cp s3://meduni-workshop-dataset-137808120700/shared/  
./download/ --recursive
```


Loading & Inspecting the Data

Show folders/data

Visualize the data (both images/metadata).

Get an idea of what the data looks like, how organized it is, and how good do you think the quality of the data is.

Data Quality Assessment & Cleaning

Identify issues *before* cleaning: missing values, duplicates, inconsistent types, outliers, encoding problems.

Understand your task: Lesion segmentation in dermatoscopic images.

1. What have you noticed about the images?
2. What issues are you expecting to arise?
3. What do you consider to be basic requirements each image + metadata pair?

Data Transformation & Standardization

Reshape, rename, or map values to match a common schema or controlled vocabulary.

Does your data need to be ordered? Renamed? What set up are you planning for inputting your data?

Data Splits

We need 3 subsets: train, validation & test.

Be careful with what **unit** you use to do the split. Would you divide images, lesions or people? What issues do you see with each of these?

Hint: Consider stratification for more representative splits.

Saving the Curated Dataset

Save your new curated dataset and properly identify it.

Document any changes.

Segmentation

Task 2

Exercise #1: Train lesion segmentator

Train a model that provides a **segmentation mask for dermoscopic images based on the provided data**. Make sure to provide relevant performance metrics for the validation and test set.

Target segmentation Dice Score: above 80%

Suggested architectures: U-Net, ResNet(PyTorch/segmentation_models_pytorch), others

Suggested libraries: PyTorch

Report Generation

Task 3



Generating lesion reports with VLM

- Use an off-the-shelf medical Vision-Language Model to generate short, structured, clinical-style reports for the dermoscopic images in your curated dataset.
- Investigate how far prompt engineering alone can push report quality, without any model fine-tuning.

Suggested techniques: few-shot prompting, chain-of-thought prompting, structured-output prompting (individually or combined)



You won't be able to access your AWS account after Wednesday noon (12 pm)